Chapter 11 Exercise: From Classes to Modules

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1 Build the Module

Create a Python file named

`complex_number.py`

Begin with the class from the warm-up:

```
class ComplexNumber:
    def __init__(self, real, imag):
        self.real = real
        self.imag = imag
    def __add__(self, other):
        return ComplexNumber(
            self.real + other.real,
            self.imag + other.imag)
    def __sub__(self, other):
        return ComplexNumber(
            self.real - other.real,
            self.imag - other.imag)
```

Each object represents

$$z = a + bi.$$

2 Complex Multiplication

For

$$z_1 = a + bi, \qquad z_2 = c + di,$$

multiplication gives

$$z_1 z_2 = (ac - bd) + (ad + bc)i.$$

Add this method:

```
def __mul__(self, other):
    real = self.real*other.real \
         - self.imag*other.imag
    imag = self.real*other.imag \
         + self.imag*other.real
    return ComplexNumber(real, imag)
```

3 Complex Conjugation

The **complex conjugate** changes the sign of the imaginary part:

$$z = a + bi \qquad \Longrightarrow \qquad z^* = a - bi.$$

Add

```
def conjugate(self):
    return ComplexNumber(
        self.real, -self.imag)
```

4 Complex Division

To divide complex numbers, use the following formula:

$$\frac{a+bi}{c+di} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}i.$$

Add

```
def __truediv__(self, other):
    den = other.real**2 \
        + other.imag**2
    real = (self.real*other.real \
          + self.imag*other.imag)/den
    imag = (self.imag*other.real \
          - self.real*other.imag)/den
    return ComplexNumber(real, imag)
```

5 Create `main.py`

In a second file, create

```
from complex_number import ComplexNumber
z1 = ComplexNumber(3, 2)
z2 = ComplexNumber(5, -4)
z_add = z1 + z2
z_sub = z1 - z2
z_mul = z1 * z2
z_div = z1 / z2
z_conj = z1.conjugate()
```

Print the real and imaginary parts:

```
print("Add:",
      z_add.real, z_add.imag)
print("Subtract:",
      z_sub.real, z_sub.imag)
print("Multiply:",
      z_mul.real, z_mul.imag)
print("Divide:",
      z_div.real, z_div.imag)
print("Conjugate:",
      z_conj.real, z_conj.imag)
```

Compare your hand calculations with the values printed by `main.py`.